Aadhithya Bharathi A

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Summary

Final-year CS undergraduate at **Amrita Vishwa Vidyapeetham** (CGPA 9.01/10) pursuing a parallel **Diploma in Data Science from IIT Madras**, with hands-on experience building production-grade full-stack, ML, and distributed systems. Proficient in end-to-end ML pipelines, Kubernetes-based observability, REST microservices, and GenAI tooling. Published applied AI research and delivered cross-platform mobile applications in a live agile environment. Seeking roles in Software Engineering, AI/ML Engineering, or Data Analytics where I can contribute technical depth from day one.

Education

Amrita Vishwa Vidyapeetham — Coimbatore, India
B.Tech Computer Science Engineering | *CGPA:* ***9.01 / 10*** — *2023 – 2027*

Indian Institute of Technology (IIT) Madras — Remote
Diploma in Data Science — *2023 – 2026*

Experience

AI Developer Intern — Jul 2026 – Oct 2026
ABB — Marine Team — *Bangalore, India (On-site)*

- Added **API versioning** to a backend services (Messaging Web API) in **.NET**, so existing clients keep working while new versions roll out; retired versions return a clear error and deprecated ones return a warning.
- Debugged a **race condition** in the edge-to-cloud messaging flow (**MQTT/AMQP**) where messages failed because they arrived before their conversation was saved; removed incorrect resend logic and added automatic retries (10 attempts, 10 s apart) with a failure notification.
- Added monitoring to the messaging service using **OpenTelemetry** and built a **Grafana** dashboard for request volume, latency and errors; improved logs with trace IDs so a single message can be followed end to end.
- Wrote **Python/pytest** automated tests that simulate outages and restart **Kubernetes** pods to check services recover, made **Playwright** UI test suites run on any environment, and analyzed failures from test pipelines.

Full Stack Developer Intern — May 2025 – Jun 2025
Trusty Bytes — *India (On-site)*

- Engineered **11 production-ready screens** in React Native (Android & iOS) for a demand-supply workflow application, delivering zero rollback post-deployment across both platforms.
- Architected a **reusable component library** that reduced UI duplication by an estimated **30%**, cutting iteration time for the 5-person engineering team.
- Operated within an agile sprint cycle, shipping across two consecutive sprints with daily standups and PR-based code reviews.

Projects

EdgeMind – Multi-Agent AI for Industrial Infrastructure Monitoring — GitHub
React Native, Kubernetes, Prometheus, LLMs

- Built for the ABB Hackathon. Developed a multi-agent AI system that monitors industrial Kubernetes workloads using Prometheus, detects cascading failures, and performs AI-powered root cause analysis with DMD-based predictive analytics.
- Designed and developed the mobile application, converting a six-page real-time dashboard into a mobile-first experience with live WebSocket updates, AI-generated root cause analysis, and predictive failure alerts.

Smart Cafeteria Management System — GitHub
Go, Flask, React, Random Forest, Docker, JWT

- Designed and deployed a containerized microservices architecture for real-time cafeteria demand forecasting with JWT authentication and two-factor authentication (2FA).
- Trained a Random Forest model on 1,800+ records, achieving approximately 81.5% forecasting accuracy while maintaining uninterrupted service through a rule-based fallback engine.

FX Decision Recommendation System GitHub
Python, Prophet, Pandas, REST API

- Developed an FX analytics platform processing over 2,500 data points per currency across ten years to generate accurate 30-day exchange-rate forecasts.
- Implemented a Value-at-Risk (VaR) engine recommending hold, partial conversion, or full conversion strategies for treasury risk management.

Pharmacy Inventory Optimization & Demand Forecasting GitHub
Python, Pandas, Time-Series Forecasting — IIT Madras Capstone

- Applied ABC and XYZ inventory classification to develop a nine-cell procurement control matrix for a live pharmacy.
- Compared multiple forecasting models, with a two-month Simple Moving Average achieving a MAPE of 2.2% on the highest-selling product.
- Identified approximately 52,838 in expiry losses and 2,695 in stock-out losses, proposing an EOQ/JIT procurement strategy to reduce inventory costs.

Technical Skills

Languages: Python, C#, JavaScript, Java, Go, SQL
Backend & APIs: .NET (ASP.NET Core), REST APIs, API Versioning, Swagger/OpenAPI, FastAPI, Flask, WebSockets, Microservices, JWT, 2FA
Messaging & Observability: MQTT, AMQP, OpenTelemetry, Grafana, Prometheus
Cloud & DevOps: Kubernetes (k3s/k3d), Docker, Helm, Azure DevOps, Azure Pipelines, Git, Redis
Testing: pytest, Playwright, Allure, Integration Testing, Test Automation
AI / ML: Scikit-learn, Pandas, NumPy, Prophet, LSTM, CNN, YOLO, LLM APIs, Prompt Engineering, Agentic / Tool-Calling Orchestration
Web & Mobile: React, React Native (Expo), HTML/CSS
Databases: MongoDB, MySQL, PostgreSQL
Data Visualization: Tableau, Matplotlib, Seaborn

Achievements

Top 6 Finalist – ABB Accelerator 2026
Selected among the **Top 6 teams** from **1,600+ teams**, **14,000+ participants**, and **300+ colleges** nationwide. Presented **EdgeMind**, an AI-powered Kubernetes observability and predictive monitoring solution, at the **ABB Innovation Centre, Bengaluru**.

Research

Multimodal AI Framework for Early Diagnosis of Fibromyalgia
Paper Presentation — Invente'25, SSN College of Engineering *Sep 2025*

- Designed a multimodal deep learning framework combining imaging and clinical tabular data for early-stage fibromyalgia detection, presented to an inter-institutional technical audience.

Certifications

Data Analytics Job Simulation Mar 2026
Deloitte Australia (Forage)

Introduction to Generative AI | Introduction to NLP | Prompt Engineering May 2026
Infosys Springboard

Extra-Curricular

Editor — ASCII Technical Newsletter 2024 – Present
Amrita Vishwa Vidyapeetham

- Curate and edit technical content for a student audience, driving clarity and engagement on CS and AI topics.

Photojournalist — Ananda Vikatan 2024 – 2025
Leading Tamil Publication

- Published documentary photo stories in one of India's highest-circulation Tamil magazines.